



Chronic resistance training: is it time to rethink the time course of neural contributions to strength gain?

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Abstract

Resistance training enhances muscular force due to a combination of neural plasticity and muscle hypertrophy. It has been well documented that the increase in strength over the first few weeks of resistance training (i.e. acute) has a strong underlying neural component and further enhancement in strength with long-term (i.e. chronic) resistance training is due to muscle hypertrophy. For obvious reasons, collecting long-term data on how chronic-resistance training affects the nervous system not feasible. As a result, the effect of chronic-resistance training on neural plasticity is less understood and has not received systematic exploration. Thus, the aim of this review is to provide rationale for investigating neural plasticity beyond acute-resistance training. We use cross-sectional work to highlight neural plasticity that occurs with chronic-resistance training at sites from the brain to spinal cord. Specifically, intra-cortical circuitry and the spinal motoneuron seem to be key sites for this plasticity. We then urge the need to further investigate the differential effects of acute versus chronic-resistance training on neural plasticity, and the role of this plasticity in increased strength. Such investigations may help in providing a clearer definition of the continuum of acute and chronic-resistance training, how the nervous system is altered during this continuum and the causative role of neural plasticity in changes in strength over the continuum of resistance training.

Keywords Brain · Spinal cord · Motoneuron · Hypertrophy · Exercise · Transcranial magnetic stimulation

Abbreviations

CMEP	Cervicomedullary motor-evoked potential
EMG	Electromyography
GABA	Gamma aminobutyric acid
H-reflex	Hoffmann Reflex
MEP	Motor-evoked potential

Mmax	Maximal compound muscle action potential
MVC	Maximal voluntary contraction
SICI	Short interval intra-cortical inhibition
TMES	Transmastoid electrical stimulation
TMS	Transcranial magnetic stimulation

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“It is probably important to pay attention to the nervous system and specificity even in advanced strength training”—Digby G. Sale’s concluding sentence in his influential review entitled “Neural adaptation to resistance training” (1988).

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Introduction

Hellebrandt (1958) first introduced the idea that motor learning was most important for rapid increases in strength, and that increases in muscular size became increasingly important with time. This notion, that neural adaptations underlie strength gain during the first few weeks of resistance training, was then shown by Moritani and deVries (1979) and subsequently discussed in an influential review by Sale (1988). They postulated that, since there were measurable increases in voluntary strength with resistance training, in the absence of changes in muscle size, neuromuscular adaptations in short-term training are dominated by adaptations in the central nervous system. However, it is important to note that, since then, skeletal muscle protein adaptations have been shown within the first 2–4 weeks of resistance

training (Staron et al. 1994). Nonetheless, Sale's train of thought has provoked many studies over the past three decades, yet the last sentence in his influential review, stated above, has largely gone neglected. This statement advocated that although neural adaptation predominates the initial increases in strength as a result of resistance training, we should not neglect the adaptations in the nervous system with advanced resistance training.

Herein, we provide a brief contextual background of the neural adaptations that have been identified to underlie increases in strength with acute-resistance training. We then discuss evidence of adaptations along the primary motor pathway (i.e. the corticospinal pathway), including evidence from motor units, in chronically resistance-trained individuals. This evidence illustrates the need for us to examine more long-term contributions from the nervous system that may mediate increases in strength with advanced training.

Differentiating acute vs. chronic-resistance training

Most literature has focused on neural adaptations from acute-resistance exercise (within a session) or training (less than 12 weeks). By studying the chronic effect of resistance training we could gain a better appreciation of the long-term benefits offered by resistance training. A clear delineation between acute and chronic-resistance training has not been well established, nor have the continued neural adaptations been quantified. In one review, for example, short-term resistance training was considered to be ≤ 6 weeks, whereas medium and long terms were defined as 7–23 and ≥ 23 weeks, respectively (Ashton et al. 2020). At present, the terms “long-term” and “chronic” resistance training are arbitrary and, thus, require more clarity. Indeed, the time course of neural adaptations to resistance training is not dichotomous, but rather there is likely a continuum. We propose that consistent training for longer than 1 year is required in order for an individual to be chronically resistance-trained. Even though this timeframe is somewhat arbitrary, we believe that this would ensure that studies have some sort of consistency when discussing the neural adaptations to chronic-resistance training.

In the above-mentioned seminal review, Digby Sale (1988) pointed out that resistance training research studies typically involve training programs that are 8–20 weeks, yet serious athletes have training programs that last from months up to years. To this day, it is commonly accepted, and appears in many reviews and textbooks, that most of the strength gain within the first 2–4 weeks of resistance training relies primarily on neural adaptation, whereas further adaptation is limited to muscular adaptation (i.e. hypertrophy). This assertion was supported by a 2-yearlong study

that examined neuromuscular adaptations to resistance training in elite weightlifters. Bipolar surface electromyography (EMG) of multiple quadriceps muscles was assessed at 4-month intervals for 24 months, and no significant changes in the amplitude of EMG activity were evident throughout the training protocol, yet the lean muscle mass did moderately increase (Häkkinen et al. 1988). However, not only was this study limited to elite weightlifters, which would allow for a narrow range of improvement, but there are serious caveats associated with using EMG amplitudes to infer changes in neural drive to the muscle (Halperin et al. 2018; Vigotsky et al. 2018).

Despite the many caveats associated with surface EMG to provide inferences about neural adaptations, Balshaw et al. (2019) recently provided a well-controlled comparison of agonist and antagonist surface EMG. They compared three groups (untrained, 12 weeks of resistance-trained, and 4 years of resistance-trained) while ensuring that the EMG was normalized and corrected for the distance between the recording electrodes and the muscle. EMG activity of the agonist was higher in both trained groups compared to the untrained group, but there was no difference between the 12 weeks and 4 years of resistance (Balshaw et al. 2018). The authors ascribed the difference in strength between the two trained groups to result from differences in muscle size. Although the quality and quantity of muscle mass certainly contribute to additional improvements in strength, lack of change in EMG amplitudes of the agonist muscle alone should not exclude further investigation of neural adaptations to chronic-resistance training. Indeed, the authors also noted that antagonist muscle activity was significantly reduced in the 4 years training group, suggesting that the neural control of the musculature required to perform the task was more efficient (i.e. net forces in the optimal direction). This falls in line with Sale's opinion, in which he believed that the expression of voluntary strength ought to be thought of as “skilled act” that would be improved if activation of prime movers was fully achieved, whereas synergists and antagonist muscles were “activated appropriately”.

Neural adaptations to resistance training throughout the years

Acute neural adaptations to resistance training have been studied extensively (see Skarabot et al. (2021) or Aagaard et al. (2020) for contemporary reviews). However, some authors have recently brought the functional relevance of training-induced neuroplasticity into question. Hortobagyi et al. (2021) make a compelling and reasonable argument that advanced statistical and study design (i.e. causal mediation analysis) should be used to determine how neural adaptations actually contribute to increases in strength.

Nonetheless, several neural adaptations have been shown in response to resistance training and we will briefly summarize them, from motoneuron to motor cortex. Figure 1 illustrates proposed central nervous system sites that undergo plasticity with resistance training and how the plasticity of these sites may contribute to changes in strength when an individual acutely and chronically resistance trains.

Increased neural activation is believed to contribute to increased strength as a result of resistance training, but neural adaptations are better appreciated when examined beyond the amplitude of surface EMG. For instance, by tracking motor units longitudinally throughout a 4-week resistance training intervention, Del Vecchio et al. (2019) showed that motor units are recruited more easily and discharge at higher rates after training. After hyperpolarization duration, estimated through the interval death rate analysis method, is also reduced in response to resistance training (Christie and Kamen 2010), suggesting that the excitability of motoneurons is increased in response to resistance training. Indeed, biophysical adaptations also occur in both low- and high-threshold motoneurons of strength trained

rodents (Krutki et al. 2017). Specifically, strength training causes increases in input resistance and reductions in rheobase, both of which indicate adaptations to membrane conductance and help explain why motor units are more easily recruited after strength training. Strength training also increases the gain of the frequency-current relationship of motoneurons, which means that the output of the motoneuron is more sensitive to small changes in inputs and helps explain the increase in firing frequencies after strength training. The number of excitatory, but not inhibitory, synapses on spinal motoneurons is also increased in a rodent model of resistance training, suggesting that they receive greater excitatory synaptic input following training (Adkins et al. 2006). These motor unit adaptations could go unnoticed if we constrained ourselves to bipolar surface EMG due to a number of both physiological and non-physiological factors (Farina et al. 2016). Beyond the motor unit, there is a need to understand alterations in the synaptic and metabotropic inputs to the motoneurons in response to resistance training.

Although it is unclear whether motoneurons continue to adapt once hypertrophy begins, advances in technology

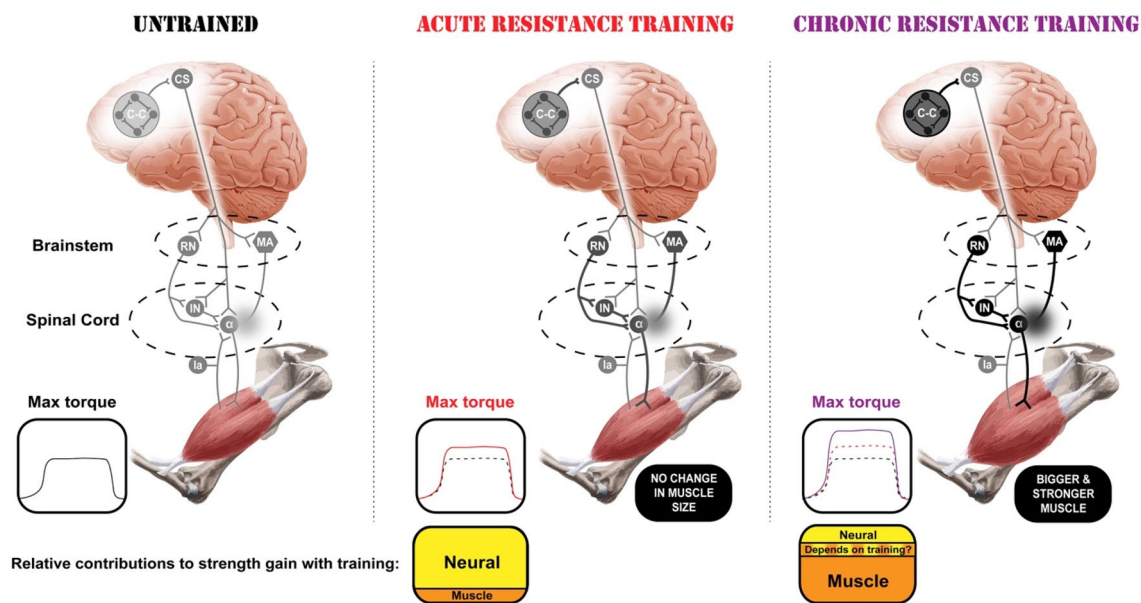


Fig. 1 A simplistic schematic illustrating the possible sites of neural adaptation with acute (middle panel)- and chronic (right panel)-resistance training. We have highlighted the sites most likely to contribute; intraspinal circuits or cortico-cortico neurons (labeled C-C), ionotropic projections of reticulospinal neurons (labeled RN), indirect pathways innervating alpha-motoneurons via interneurons (labeled IN), intrinsic alpha motoneuron properties (labeled α), or metabotropic reticulospinal projections that provide monoaminergic drive to the spinal neurons (labeled MA). Sites that are less likely to contribute to increases in strength with resistance training are Ia afferents (labeled Ia), and the corticospinal neurons (labeled CS). From untrained to acute-resistance training, one can notice that the sites most likely to adapt become darker and more prominent. Likewise,

from acute training to chronic training, we speculate that similar sites are most likely to adapt. These adaptations accompany changes in max torque, which is depicted in the inset. Below the inset, we provide a relative contribution of muscular (orange) and neural (yellow) adaptations that contribute to strength gain. It is important to note that neither acute nor chronic training rely solely on muscular or neural adaptations, but rather there is a shift from predominantly neural during acute training to predominantly muscular with chronic training. Most importantly, we speculate that there is almost certainly some portion of neural contribution to the increase in strength with chronic training, and these contributions likely vary based on the training performed (indicated by the hatched yellow/orange fill)

may allow for inferences about these adaptations in the near future (Farina et al. 2016). Despite these advances in technology that will allow us to track motor units over time (Vecchio and Farina 2019), the cause of the reduced recruitment thresholds and increased firing rates is largely unknown, but synaptic input from various sources likely contribute. It is currently unclear whether afferent feedback is altered with resistance training. For instance, it has been speculated many times that Ib inhibition or Golgi tendon organ (GTO) sensitivity is altered with resistance training, but there is currently a lack of empirical evidence to support such notion (see Chalmers 2002 for discussion on the topic). Contrasting results between H-reflex studies (Lagerquist et al. 2006; Del Balso and Cafarelli 2007) also suggest that Ia reflex adaptations to resistance training are not entirely clear. Spinal circuitry affecting H-reflex excitability appears to change in a more concise fashion in response to resistance training. Young people with a long history of power training exhibit reduced group Ia presynaptic inhibition (Koceja et al. 2004), and 4 weeks of tibialis anterior resistance training increases Ia disynaptic reciprocal inhibition from the common peroneal nerve to the soleus H-reflex (Geertsen et al. 2008). These alterations in spinal circuitry may help explain reductions in co-contraction reported in Balshaw et al. (2019) and elsewhere (i.e. Pearcey et al. (2014)), but are unlikely to account for all of the increase in strength in response to resistance training since they are mediated by descending input.

Evidence for reorganization of the movement representation area of the motor cortex in response to strength training comes from a study in rodents that were trained to reach for and break noodles of increasing size (Remple et al. 2001). This reorganization is similar to rodents who train the same novel reaching task that does not involve progressive increases in strength, suggesting that some neural adaptations in the brain may resemble learning a new movement pattern, and others may be specific to improvements in neural drive to the muscle. Farthing et al. (2007) measured changes in blood oxygen level-dependent (i.e. BOLD) signal in response to 6 weeks of strength training and showed widespread changes in the “neural activation” patterns after training in premotor and motor areas. Using diffusion tensor imaging, Palmer et al. (2013) also showed that there are structural changes to the corticospinal tract in response to 4 weeks of strength training. Their results showed a decreased diffusivity, which indicates an increase in the myelination of corticospinal tract axons involved in activation of the trained muscle that improve efficiency of transmission from the motor areas of the brain to the spinal motoneurons. These structural changes in the motor areas of the brain and the corticospinal tract would act to improve efficiency of signals to motor units, and warrant the many studies discussed below that have been performed to

examine corticospinal pathway adaptations in response to resistance training.

Measuring output from the corticospinal pathway in humans

The corticospinal tract is one of the main descending pathways contributing to the transmission of voluntary central commands leading to activation of spinal motoneurons and subsequently, the activation of skeletal muscle. Given its importance in voluntary control of movement, the excitability of the corticospinal pathway is commonly examined to infer whether adaptations occur along this pathway in response to resistance training through the use of non-invasive stimulation paradigms, such as transcranial magnetic stimulation (TMS) and transmastoid electrical stimulation (TMES). Excellent reviews have recently been published on these techniques (TMS: Kalmar (2018); TMES: McNeil et al. (2013)), thus we will provide a brief description herein.

Single pulse TMS applied over the motor cortex activates upper motoneurons transynaptically, resulting in a MEP that is recorded via EMG from the muscle of interest. The peak-to-peak amplitude of the MEP is used as a measure of the excitability along the entirety of the corticospinal pathway. The single-pulse MEP recorded at the muscle is a gross response that is shaped by neurons and their excitability in the (1) brain, including cortico–cortico and cortico-motoneuronal neurons, (2) spinal cord, including interneurons and alpha-motoneurons, and (3) periphery, including the neuromuscular junction and muscle. To help elucidate the loci (supraspinal or spinal) and mechanisms underlying modulation of the MEP, additional techniques have been developed. Paired-pulse TMS paradigms are used to examine the excitability of various cortical circuits that may influence overall corticospinal excitability, though a detailed description of all of the techniques and physiology of these circuits is beyond the scope of the current review. However, we do discuss the effect of chronic-resistance training on short interval intra-cortical inhibition (SICI) below.

TMES is applied at the level of the mastoid processes and produces cervicomedullary motor-evoked potentials (CMEPs) which are used as a measure of spinal excitability. Importantly, CMEPs are considered the most direct method to test motoneuron excitability in response to synaptic input (McNeil 2013), have a large monosynaptic component, are not influenced by changes in cortical excitability, and are probably not influenced by presynaptic inhibition on many of the descending pathways (Nielsen and Petersen 1994). These techniques are now commonly used in combination to assess changes in supraspinal and spinal excitability, but both MEP and CMEP responses are susceptible to changes in excitability of the periphery. Thus, to account for the

influence of the peripheral excitability on these responses, a maximal compound muscle action potential ($M_{\max}$, i.e. the summation of the electrical activity of the motor units activated by the electrical stimulus (Rodriguez-Falces and Place 2018) is evoked via peripheral nerve stimulation. The MEP and CMEP can then be normalized to the $M_{\max}$ to illustrate whether or not that any change in the MEP or CMEP response is due to peripheral factors.

Adaptations along the corticospinal pathway following acute-resistance training

Extensive reviews have recently been published regarding adaptations of the central nervous system to acute (i.e. multiple sessions over a 3–12 weeks) resistance training (Skarabot et al. 2021; Hortobagyi et al. 2021; Aagaard et al. 2020; Siddique et al. 2020b). Other reviews have focused specifically on plasticity of corticospinal pathway to single day resistance exercise (not discussed here) (Mason et al. 2019) and acute-resistance training (i.e. multiple sessions over a 3–12 weeks) (Tallent et al. 2021; Kidgell et al. 2017). Here, we will briefly summarize the adaptations along the corticospinal pathway in response to acute-resistance training. Carroll et al. (2002) first showed that 4-week resistance training increased index finger abduction strength while reducing the MEP amplitude during 40–60% of maximal voluntary contraction (MVC). Since then, many acute-resistance training studies have studied if various segments of the corticospinal pathway undergo plasticity in response to acute-resistance training. Overall, the research has been unequivocal with regards to changes in MEP amplitude [i.e. see Fig. 1 middle panel where the corticospinal (CS) neurons are unaltered with acute-resistance training], which, as mentioned previously, is a gross measure of excitability of the entire corticospinal pathway. Discrepancies in findings may be due to the training paradigm applied, the task at hand, stimulation paradigm, whether or not the MEP is recorded at rest or during a muscle contraction, the muscle studied, and/or changes occurring at other segments along the corticospinal pathway (Kidgell et al. 2017; Siddique et al. 2020a, b). Based on CMEP studies (Nuzzo et al. 2017; Siddique et al. 2020b), and similar to H-reflex studies, it appears that the spinal motoneurons do not respond differently to single-pulse inputs following acute-resistance training, despite evidence to suggest more excitable motoneurons during repetitive firing. Finally, following acute-resistance training, intra-cortical inhibition is reduced, which is evidenced by decreased corticospinal silent periods [mediated by the neurotransmitter gamma-aminobutyric acid-B ($GABA_B$)] (Siddique et al. 2020b) and decreased SICI (the activation of the intra-cortical inhibitory $GABA_A$ network (Weier et al. 2012); see Fig. 1 middle panel where the cortico-cortico

[C–C] circuits becomes more pronounced with acute-resistance training).

Adaptations along the corticospinal pathway with chronic-resistance training

Only a handful of studies have paid attention to neural adaptations in chronically resistance-trained individuals. Collecting longitudinal data from retention and methodological stand points are quite difficult, but longitudinal studies would help identify if chronic-resistance training leads to continued neuroplasticity from that seen in response to acute-resistance training. Cross-sectional designs allow for the comparison of individuals who are already chronically resistance-trained with those who are non-resistance-trained, which can provide insights about resistance training-induced neural plasticity. del Olmo et al. (2006) compared TMS-evoked responses between 10 individuals with more than 2 years of resistance training experience and 10 non-resistance-trained participants. They evoked MEPs with 100% of maximum stimulator output in the biceps brachii while participants completed a series of isometric contractions ranging from 10 to 90% of MVC, and reported no difference in MEP amplitude between the groups. Tallent et al. (2013) also found that MEP amplitudes recorded from the tibialis anterior did not differ between chronic-resistance-trained (> 3 years) and non-resistance-trained groups when isometric, concentric and eccentric contractions were performed. However, the lack of reported change in corticospinal excitability in their study may be attributed to the selected muscle (i.e. tibialis anterior) that is rarely targeted in the majority of resistance training programs (Griffin and Cafarelli 2007) compared to the other muscles of the body (i.e. biceps brachii or quadriceps). In addition, the two aforementioned studies only measured MEP amplitude, which as stated above, is limited as a measure of gross corticospinal excitability without differentiating between supraspinal and spinal adaptations.

More recently, our laboratory combined TMS with TMES to investigate plasticity along the corticospinal pathway in chronic-resistance-trained people. We showed a reduction in MEP but not CMEP amplitudes recorded from the biceps brachii for chronic-resistance-trained (≥ 2 years) individuals between 50 and 100% of elbow flexor MVC compared to recreationally active (non-resistance-trained) individuals (Pearcey et al. 2014). The decreased MEP with no change in CMEP would indicate that chronic-resistance-trained individuals have neuroplasticity of the corticospinal pathway upstream of the spinal motoneuron. However, we speculated that increased firing rates of the spinal motoneuron (Matthews 1999; Olivier et al. 1995) may “mask” the MEP response. A year later, Philpott et al. (2015) showed that

chronic-resistance-trained (≥ 2 years) individuals had greater CMEP amplitudes in the non-dominant biceps brachii than non-resistance-trained individuals during 50% and 75% of elbow flexor MVC, indicating differences in motoneuron output between the two groups. The effects of chronic-resistance training on biophysical and rhythmic motoneuron properties comes from the previously mentioned work of Krutki et al. (2017) who showed that rats that were resistance-trained for 5 weeks (which equates ~ 3 years in human biological time) had more excitable motoneurons (predominantly seen in fast motoneurons). In particular, these motoneurons had higher firing frequencies per given amount of current injection and had higher steady-state firing frequencies. This motoneuron plasticity would allow for enhanced motor unit force production across all contraction intensities as a result of resistance training. Indeed, Pearcey et al. (2014) clearly demonstrated that force production across all contraction intensities (10–100% MVC) were higher in chronic-resistance-trained individuals and that the CMEP/force ratio was also lower across all intensities.

In addition to intrinsic properties of the spinal motoneurons, chronic-resistance training may alter metabotropic (i.e. monoaminergic) input originating from brainstem nuclei. Both inputs could alter persistent inward currents (Heckman et al. 2004) and enhance the excitability of the motoneuron and subsequently increase firing rates, especially at higher contraction intensities. Adaptations in the reticulospinal tract in the absence of corticospinal contributions has been shown in non-human primates following a ~ 3 month resistance training intervention (Glover and Baker 2020). The reticulospinal tract has both ionotropic and metabotropic projections (see Fig. 1), with the latter being potent facilitators for exciting motoneurons via activation of previously mentioned persistent inward currents (Heckman et al. 2008). The aforementioned studies support the notion that chronic-resistance training enhances how the spinal motoneuron responds to input and generates its output, leading to enhanced force. Thus, spinal motoneuron plasticity and adaptations within the reticulospinal pathway may contribute to chronic-resistance training-induced alterations observed along the corticospinal pathway (see Fig. 1 right panel where the alpha motoneuron [α], monoaminergic input from the brainstem [MA], and ionotropic reticulospinal [RS] neurons are more pronounced with chronic-resistance training).

Plasticity of intra-cortical circuits may also occur with chronic-resistance training [see Fig. 1 cortico-cortico (C–C) circuit]. To further probe specific cortical circuitry that cannot be achieved with the use of single-pulse TMS, we became interested in chronic-resistance training-induced changes in SICI. To our knowledge, only one study, which is from our laboratory, has examined how intra-cortical circuits interact with corticospinal neurons in individuals who are chronically resistance-trained.

Lahouti et al. (2019) compared active corticospinal motor threshold and SICI of the biceps brachii measured during 15, 25 and 40% MVC of the elbow flexors between individuals with more than 2 years of continuous resistance training and individuals with no history resistance training. The hypothesis was simple; reductions in inhibition within the cortex of the chronic-resistance training group would show a reduced active motor threshold and SICI compared to the non-resistance-trained group. These factors would manifest as a greater facilitation of the cortical neurons during different intensity contractions. Both the active motor threshold and SICI (i.e. reduced inhibition leading to facilitation) were reduced during 25 and 40% MVC of the elbow flexors in the chronic-resistance-trained, compared to the non-resistance-trained group. Based on the results, we suggested that the mechanism underpinning the differences between the groups was differential activation of intra-cortical networks. These networks were altered in a way that would facilitate strong muscle contractions, thus producing higher force output without requiring greater descending input from the motor cortex. Recent work from Glover and Baker (2020) also provides evidence for neural adaptations in intra-cortical circuits after 3 months of resistance training in primates. Therefore, another neural adaptation along the primary motor pathway with chronic-resistance training is at the supraspinal level involving the neural circuitry for inhibiting and facilitating corticospinal neurons. Future studies that include further measures of inhibition and facilitation could help to determine further mechanisms underpinning plasticity of intra-cortical circuitry with chronic-resistance training.

Based on the limited studies detailing the effects of chronic-resistance training on the corticospinal pathway, it can be surmised that plasticity along the corticospinal pathway occurs (1) spinally—at the alpha motoneuron that leads to greater firing frequencies of the motor units, and (2) supraspinally—including intra-cortical circuitry that leads to reduced inhibition or facilitation of the upper motoneurons. We have provided a graphical summary of the hypothesized sites of neural adaptations based on work in acute-resistance training studies, and limited evidence in chronic-resistance training studies, in Fig. 1. Both spinal and supraspinal adaptations, in part, allow for greater force generation and, probably, more efficient nervous system during this force generation in chronically resistance-trained individuals. For now, it appears that chronic-resistance training has no effect on gross corticospinal excitability, at least in the pathway from the cortex to the biceps brachii. Although more work is needed to confirm this statement, we hope that this call to action can help inspire progress towards understanding neural adaptations to chronic-resistance training.

Often overlooked complexities of “real-world” resistance training

Resistance training in a lab environment rarely equates to the training actually performed in the “real world” by elite athletes, weekend warriors, or—anyone for that matter! Effective resistance training programs are heterogeneous because they are highly individualized [i.e. based on the goals of an individual (Fleck and Kraemer 1997)] and progressive in nature (Kraemer and Ratamess 2004). Non-specific resistance training [i.e. the Delorme method of 3 sets of 10 repetitions (Todd et al. 2012)] can be effective at increasing strength in untrained or novice individuals. Advanced training, on the other hand, involves systematic variation of acute training variables, such as the exercise selection (unilateral vs. bilateral, single joint vs. multi-joint), muscle action (i.e. concentric, eccentric, or isometric), intensity (i.e. load or percent exertion), volume (number of reps and sets), mode of exercise (i.e. machine vs. free-weights), order of exercises, rest periods, velocity of training, and frequency of training sessions, which all contribute to optimize progress towards predetermined goals (American College of Sports 2009). Rather than completing the same training intervention over and over, effective resistance training maximizes progression through progressive overload, specificity of training, and variation in the training program (Kraemer and Ratamess 2004). For instance, when considering the “strength-endurance continuum,” higher intensities are used to effectively build strength, whereas higher numbers of repetitions at lower intensities are used to effectively build muscular endurance (Campos et al. 2002). Even when the above-mentioned factors are considered in program design, they are tough to track, both in terms of dosage of exercise completed, and the individual contributions of each training variable to neural adaptations. Thus, the complexities of “real-world” resistance training programs (i.e. chronic-resistance training) certainly should not simply be considered a temporal extension of resistance exercise or short-term resistance training studies.

The majority of studies cited in the previous sections included strength training programs that lack the complexity of “real world” resistance training. Research from

our lab shows that there are measurable differences in the central nervous system between untrained and chronic-resistance-trained individuals. However, this research and the subsequent conclusions are based on cross-sectional study designs, lack specifics on the many aforementioned resistance-training program variables, and, as importantly, the time course of nervous system adaptations. A series of longitudinal studies focusing on the effects of various chronic-resistance training programs on the nervous system would help alleviate some of these shortcomings. Furthermore, participant recruitment would have to be well-thought-out because currently we also lack understanding of how age, sex, prior training experience, and disease/dysfunction affect the time course of nervous system adaptation during chronic-resistance training.

Is it time to reconsider the classical graph from Digby G. Sale?

Since longitudinal evidence is not available, the current understanding of neural adaptations to chronic-resistance training is limited to a few cross-sectional studies. Nonetheless, adaptations of intraspinal circuits and spinal motoneurons appear evident when comparing groups that are chronically resistance training compared to those who are not. Whether or not chronic-resistance training provides continual adaptations in the central nervous system remains an open question, but it would be naïve to assume that a well-planned resistance training program that alters training variables to induce stress on the neuromuscular system would not cause continual adaptations in the nervous system, since it appears to have an immediate onset of plasticity to any stressor. Therefore, although much work is left to be done to support the notion, we suggest that the time course of contributions from the neuromuscular system should be updated to temper the plateau in neural adaptations to resistance training that has been previously suggested by Sale 1988 (see Fig. 2). In particular, undulating adaptations in strength may be achieved irrespective of hypertrophy if the training program adequately introduces novel training stressors.

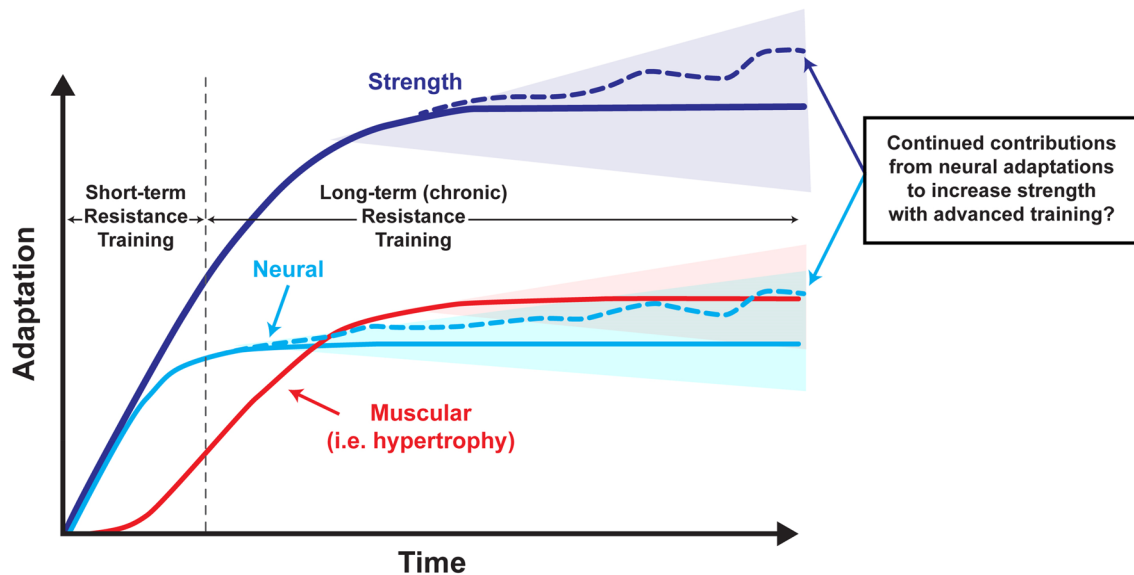


Fig. 2 An update to Digby G. Sale's summary of the relative contribution of neural (cyan) and muscular (red) adaptations to the improvement in strength (navy) as a function of time. The vertical broken line indicates the 8–12-week mark, which corresponds to the duration of time that most training studies are performed, and neural adaptation predominate. To the right, with longer-term resistance training strength gain is dominated by muscular adaptations (i.e. hypertrophy), but we speculate that there are also contributions arising

from neural adaptations, indicated by the broken lines. While the relative contribution of muscle hypertrophy to overall strength gain is shown to plateau we would argue that an increase in strength gain will also be, in part, due to contributions arising from muscle hypertrophy (however not shown here) (created with inspiration from Sale 1988). Variability in the frequency, intensity, time, and type of training will cause various neural, and muscular adaptations, and ultimately affect strength, which is indicated by the shaded regions

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

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